

A Guide for Choosing Community Detection Algorithms in Social Network Studies: The Question Alignment Approach



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Introduction: Community detection, the process of identifying subgroups of highly connected individuals within a network, is an aspect of social network analysis that is relevant but potentially underutilized in prevention research. Guidance on using community detection methods stresses aligning methods with specific research questions but lacks clear operationalization. The Question Alignment approach was developed to help address this gap and promote the high-quality use of community detection methods.

Methods: A total of 6 community detection methods are discussed: Walktrap, Edge-Betweenness, Infomap, Louvain, Label Propagation, and Spinglass. The Question Alignment approach is described and demonstrated using real-world data collected in 2013. This hypothetical case study was conducted in 2019 and focused on targeting a hand hygiene intervention to high-risk communities to prevent influenza transmission.

Results: Community detection using the Walktrap method best fit the hypothetical case study. The communities derived using the Walktrap method were quite different from communities derived through the other 5 methods in both the number of communities and individuals within communities. There was evidence to support that the Question Alignment approach can help researchers produce more useful community detection results. Compared to other methods of selecting high-risk groups, the Walktrap produced the most communities that met the hypothetical intervention requirements.

Conclusions: As prevention research incorporating social networks increases, researchers can use the Question Alignment approach to produce more theoretically meaningful results and potentially more useful results for practice. Future research should focus on assessing whether the Question Alignment approach translates into improved intervention results.

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INTRODUCTION

Social networks are related to health.^{1–3} Social network analysis has emerged as one tool for evaluating and understanding relationships between individual health and social environments as well as developing interventions to prevent disease and improve public health.^{4–6} A variety of social network analytic methods are available to describe network composition, examine relationships between social networks and health, or identify the underlying group structure of social networks.^{2,7}

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